Problem Solving (MATH 1000) **Name (Print):** ____________________
Summer 2018
Thursday, 06/14/2018
Quiz 4: Section 10.4, 13.1, 13.2

Score:	/ 100

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1. Oh, the difference a dollar makes! Suppose you run up a balance of \$10,000 on a credit card, with an interest rate of 22% compounded monthly.

   (a) $\boxed{\text{10 points}}$ If you only pay \$184.33 per month, how long will it take to pay off the balance?

$$\text{HINT: } T = \frac{\ln\left(\dfrac{M}{M - pP}\right)}{\ln(1+p)}$$

   (b) $\boxed{\text{5 points}}$ How much interest do you pay over this time span?

(c) $\boxed{\text{10 points}}$ If you only pay \$183.34 per month, how long will it take to pay off the balance?

(d) $\boxed{\text{5 points}}$ How much interest do you pay over this time span?

(e) $\boxed{\text{5 points}}$ If you only pay \$183.33 per month, how long will it take to pay off the balance? Will the balance ever be paid off?

HINT: If you end up trying to take the natural logarithm of a negative number when using the formula given in the first hint, then it means the balance will never be paid off.

2. Oh, the difference a percent makes! Suppose you run up a balance of \$10,000 on a credit card, with an interest rate of 21% compounded monthly.

   (a) $\boxed{\text{10 points}}$ If you only pay \$184.33 per month, how long will it take to pay off the balance?

(b) $\boxed{\text{5 points}}$ How much interest do you pay over this time span?

(c) $\boxed{\text{10 points}}$ If you only pay \$183.34 per month, how long will it take to pay off the balance?

(d) $\boxed{\text{5 points}}$ How much interest do you pay over this time span?

(e) $\boxed{\text{10 points}}$ If you only pay \$183.33 per month, how long will it take to pay off the balance?

3. $\boxed{\text{10 points}}$ Give the formula for the Nth Fibonacci number in terms of previous Fibonacci numbers and write the first eight numbers in the Fibonacci sequence.

4. $\boxed{\text{5 points}}$ State the golden property and write the exact value of the golden ratio (not a decimal approximation of it).

5. $\boxed{\text{10 points}}$ Using your answer from 4, write the exact value of ϕ^2 (not a decimal approximation of it).